

Bayesian model averaging

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Bayesian model averaging

Let $\{M_\gamma : \gamma \in \Gamma\}$ indicate a set of models for a particular data set y . If Δ is a quantity of interest, e.g. effect size, a future observable, or the utility of a course of action, then its posterior distribution is

$$p(\Delta|y) = \sum_{\gamma \in \Gamma} p(\Delta|M_\gamma, y)p(M_\gamma|y)$$

where

$$p(M_\gamma|y) = \frac{p(y|M_\gamma)p(M_\gamma)}{p(y)} = \frac{p(y|M_\gamma)p(M_\gamma)}{\sum_{\lambda \in \Gamma} p(y|M_\lambda)p(M_\lambda)}$$

is the **posterior model probability** and

$$p(y|M_\gamma) = \int p(y|\theta_\gamma, M_\gamma)p(\theta_\gamma|M_\gamma)d\theta_\gamma$$

is the **marginal likelihood for model M_γ** and θ_γ is the set of parameters in model M_γ .

Bayesian model averaged moments

Since $p(\Delta|y)$ is a discrete mixture, we may be interested in simplifying inference concerning Δ to a couple of moments. Let $\hat{\Delta}_\gamma = E[\Delta|y, M_\gamma]$. Then the expectation is

$$E[\Delta|y] = \sum_{\gamma \in \Gamma} \hat{\Delta}_\gamma p(M_\gamma|y)$$

and the variance is

$$\text{Var}[\Delta|y] = \left[\sum_{\gamma \in \Gamma} (\text{Var}[\Delta|y, M_\gamma] + \hat{\Delta}_\gamma^2) p(M_\gamma|y) \right] - E[\Delta|y]^2$$

The appealing aspect here is that the moments only depend on the moments from each individual model and the posterior model probability.

Difficulties with BMA

- Evaluating the summation can be difficult since $|\Gamma|$, the cardinality of Γ , might be huge.
- Calculating the marginal likelihood.
- Specifying the prior over models.
- Choosing the class of models to average over.

Reducing cardinality

If $|\Gamma|$ is small enough, we can enumerate all models and perform model averaging exactly. But if $|\Gamma|$ is too large, we will need some parsimony.

Rather than summing over Γ , we can only include those models whose posterior probability is sufficiently large

$$\mathcal{A} = \left\{ M_\gamma : \frac{\max_\lambda p(M_\lambda|y)}{p(M_\gamma|y)} = \frac{\max_\lambda p(y|M_\lambda)p(M_\lambda)}{p(y|M_\gamma)p(M_\gamma)} \leq C \right\}$$

relative to other models where C is chosen by the researcher. Also, appealing to Occam's razor, we should exclude complex models which receive less support than sub-models of that complex model, i.e.

$$\mathcal{B} = \left\{ M_\gamma : \forall M_\lambda \in \mathcal{A}, M_\lambda \subset M_\gamma, \frac{p(M_\lambda|y)}{p(M_\gamma|y)} < 1 \right\}$$

So, we typically sum over the smaller set of models $\Gamma' = \mathcal{B}$.

Searching through models

One approach is to search through models and keep a list of the best models. To speed up the search the following criteria can be used to decide what models should be kept in Γ' :

- When comparing two nested models, if a simpler model is rejected, then all submodels of the simpler model are rejected.
- When comparing two non-nested models, we calculate the ratio of posterior model probabilities

$$\frac{p(M_\gamma|y)}{p(M_{\gamma'}|y)}$$

if this quantity is less than O_L , we reject M_γ and if it is greater than O_R we reject $M_{\gamma'}$.

Using MCMC to search through models

Construct a neighborhood around $M^{(i)}$ (the current model in the chain), call it $nbh(M^{(i)})$. Now propose a draw M^* from the following proposal distribution

$$q(M^*|M^{(i)}) = \begin{cases} 0 & \forall M^* \notin nbh(M^{(i)}) \\ \frac{1}{|nbh(M^{(i)})|} & \forall M^* \in nbh(M^{(i)}) \end{cases}$$

Set $M^{(i+1)} = M^*$ with probability $\min\{1, \rho(M^{(i)}, M^*)\}$ where

$$\rho(M^{(i)}, M^*) = \frac{p(M^*|y)}{p(M^{(i)}|y)} \frac{|nbh(M^{(i)})|}{|nbh(M^*)|}$$

and otherwise set $M^{(i+1)} = M^{(i)}$. This Markov chain converges to draws from $p(M_\gamma|y)$ and therefore can estimate posterior model probabilities.

Evaluating the marginal likelihoods

Recall that as the sample size n increases, the posterior converges to a normal distribution (Bernstein von Mises Theorem). Let

$$g(\theta) = \log(p(y|\theta, M)p(\theta|M)) = \log p(y|\theta, M) + \log p(\theta|M)$$

Let $\hat{\theta}_{MAP}$ be the MAP for θ in model M . Taking a Taylor series expansion of $g(\theta)$ around $\hat{\theta}_{MAP}$, we have

$$g(\theta) \approx g(\hat{\theta}_{MAP}) - \frac{1}{2}(\theta - \hat{\theta}_{MAP})A(\theta - \hat{\theta}_{MAP})^\top$$

where A is the negative Hessian of $g(\theta)$ evaluated at $\hat{\theta}_{MAP}$. Combining this with the first equation and exponentiating, we have

$$p(y|\theta, M)p(\theta|M) \approx p(y|\hat{\theta}_{MAP}, M)p(\hat{\theta}_{MAP}) \exp\left(-\frac{1}{2}(\theta - \hat{\theta}_{MAP})A(\theta - \hat{\theta}_{MAP})^\top\right)$$

Hence, the approximation to $p(\theta|y, M) \propto p(y|\theta, M)p(\theta|M)$ is normal.

Evaluating the marginal likelihoods (cont.)

If we take the integral over θ of both sides and take the logarithm, we have

$$\log p(y|M) \approx \log p(y|\hat{\theta}_{MAP}, M) + \log p(\hat{\theta}_{MAP}|M) + \frac{p}{2} \log(2\pi) - \frac{1}{2} \log |A|$$

where p is the dimension of θ , i.e. the number of parameters. We call this approximation the **Laplace approximation**.

Another approximation that is more computationally efficient but less accurate is to only retain terms that increase with n :

- $\log p(y|\hat{\theta}, M)$ increases linearly with n
- $\log |A|$ increases as $p \log n$

As n gets large $\hat{\theta}_{MAP} \rightarrow \hat{\theta}_{MLE}$. Taking these two together we have

$$\log p(y|M) \approx \log p(y|\hat{\theta}_{MLE}, M) - \frac{p}{2} \log n$$

Multiplying by -2, we obtain Schwarz's Bayesian Information Criterion (BIC)

$$BIC = -2 \log p(y|\hat{\theta}_{MLE}, M) + p \log n$$

Priors over models

For data-based comparisons of models, you can use Bayes Factors directly since

$$BF(M_\gamma : M_{\gamma'}) = \frac{p(y|M_\gamma)}{p(y|M_{\gamma'})} = \frac{\int p(y|\theta_\gamma)p(\theta_\gamma|M_\gamma)d\theta_\gamma}{\int p(y|\theta_{\gamma'})p(\theta_{\gamma'}|M_{\gamma'})d\theta_{\gamma'}}$$

where the last equality is a reminder that priors over parameters still matter.

For model averaging, you need to calculate posterior model probabilities which require specification of the prior probability of each model. One possible prior for regression models is

$$p(M_\gamma) = \prod_{i=1}^p w_i^{1-\gamma_i} (1 - w_i)^{\gamma_i}$$

Setting $w_i = 0.5$ corresponds to a uniform prior over the model space.

BMA output

The quantities of interest from BMA are typically

- Posterior model probabilities $p(M_\gamma|y)$
- Posterior inclusions probabilities (for regression)

$$p(\text{including explanatory variable } i|y) = \sum_{\gamma \in \Gamma} p(M_\gamma|y) I(\gamma_i = 1)$$

which provides an overall assessment of whether explanatory variable i is important or not.

- Posterior distributions, means, and variances for “parameters”, e.g.

$$E[\theta_i|y] = \sum_{\gamma \in \Gamma} E[\theta_{\gamma,i}|y] p(M_\gamma|y)$$

But does this make any sense? What happened to θ_γ ?

- Predictions:

$$p(\tilde{y}|y) = \sum_{\gamma \in \Gamma} p(M_\gamma|y) p(\tilde{y}|M_\gamma, y)$$

R packages for BMA

There are two main packages for Bayesian model average in R

- BMA: glm model averaging using BIC
- BMS: lm model averaging using g-priors and (possibly) MCMC

Back up on CRAN is

- BAS: lm model averaging with a variety of priors and (possibly) MCMC (additionally performed sampling without replacement)

BMA in R

```
library(BMA)
UScrime <- MASS::UScrime

# Set up data
x = UScrime[,-16]
y = log(UScrime[,16])
x[,-2] = log(x[,-2])

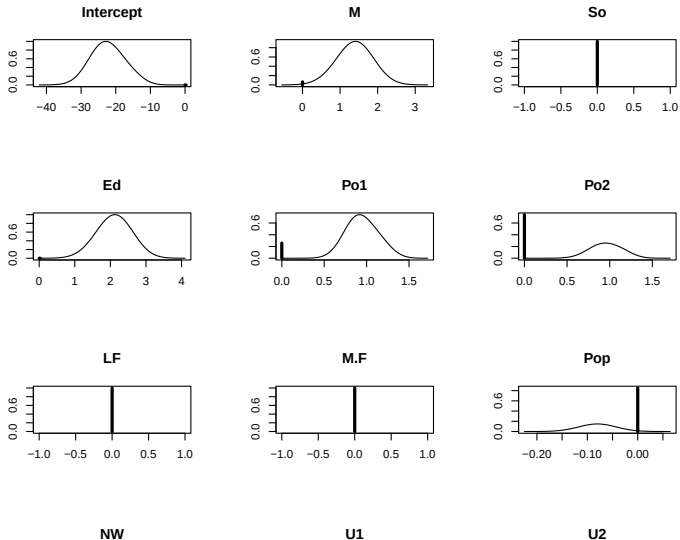
# Run BMA using BIC
lma = BMA::bicreg(x, y,
  strict = TRUE, # remove submodels that are less likely
  OR = 20)      # maximum BF ratio
```

```
summary(lma)
```

```
##
## Call:
## BMA::bicreg(x = x, y = y, strict = TRUE, OR = 20)
##
##
## 15 models were selected
## Best 5 models (cumulative posterior probability = 0.6774 ):
##
##      p!=0    EV      SD    model 1    model 2    model 3    model 4    model 5
## Intercept 100.0 -22.23688 5.10541 -22.63715 -24.38362 -24.50477 -18.26547 -26.29204
## M          93.5  1.29995 0.59146  1.47803  1.51437  1.46061  1.12775  1.68239
## So         0.0  0.00000 0.00000  .         .         .         .         .
## Ed        100.0  2.10530 0.51224  2.22117  2.38935  2.39875  1.83590  2.08662
## Po1       74.2  0.70466 0.44869  0.85244  0.91047  .         0.89054  1.14936
## Po2       25.8  0.24704 0.43075  .         .         0.90689  .         .
## LF         0.0  0.00000 0.00000  .         .         .         .         .
## M.F        0.0  0.00000 0.00000  .         .         .         .         .
## Pop        14.8 -0.01189 0.03188  .         .         .         .         .
## NW        84.1  0.08479 0.05307  0.10888  0.08456  0.08534  0.11670  .
## U1         0.0  0.00000 0.00000  .         .         .         .         .
## U2        66.3  0.20895 0.18420  0.28874  0.32169  0.32977  .         0.33564
## GDP        2.8  0.02025 0.13628  .         .         .         .         .
## Ineq      100.0  1.33551 0.32647  1.23775  1.23088  1.29370  1.24269  1.57576
## Prob      98.1 -0.23801 0.10132 -0.31040 -0.19062 -0.20614 -0.33521 -0.16699
## Time      34.1 -0.10186 0.16713 -0.28659  .         .         -0.33177  .
##
##      nVar
## r2      8      7      7      7      6
## BIC     0.842  0.826  0.823  0.821  0.805
##      -55.91243 -55.36499 -54.40788 -53.79094 -53.62430
## post prob 0.234  0.178  0.110  0.081  0.074
```

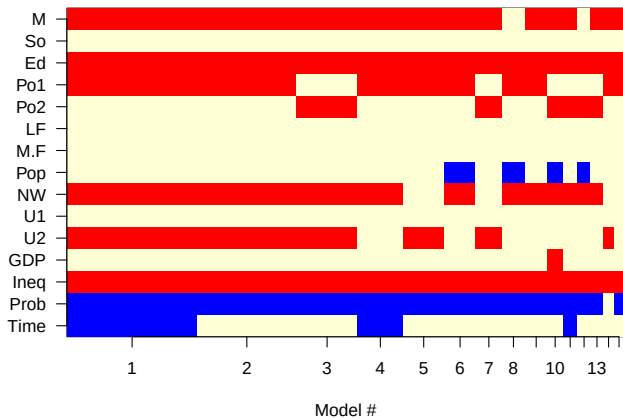
Does this make any sense?

```
plot(lma)
```



```
imageplot.bma(lma)
```

Models selected by BMA



BMA for GLMs in R

```
# Use BIC-based BMA
lma <- BMA::bic.glm(

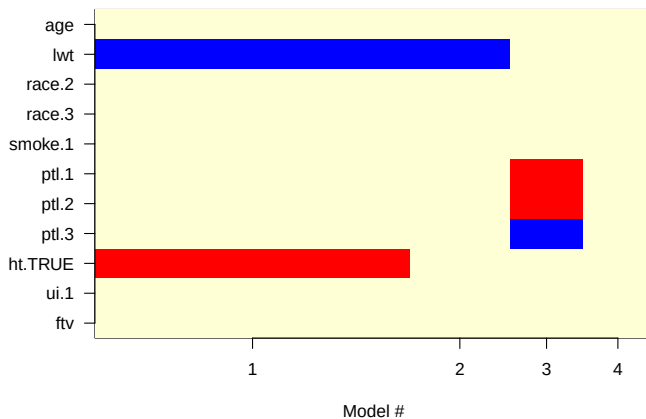
  x = MASS::birthwt |>
    dplyr::select(-low,-bwt) |>
    mutate(race = factor(race),      # mother's race (1 = white, 2 = black, 3 = other)
           smoke = factor(smoke),    # smoking status during pregnancy.
           ptl = factor(ptl),        # number of previous premature labours
           ht = factor(ht>=1),       # history of hypertension
           ui = factor(ui)),         # presence of uterine irritability,

  y = MASS::birthwt$low,             # 1 indicates birthweight < 2.5 kg,
  strict = TRUE,                     # remove models with more likely submodels
  OR = 20,                           # maximum ratio for excluding models

  glm.family = "binomial",
  factor.type = TRUE)                # include/exclude factor levels as a group
```

```
##
## Call:
## bic.glm.data.frame(x = mutate(dplyr::select(MASS::birthwt, -low,      -bwt), race = factor(race), smoke = fac
##
##
## 4 models were selected
## Best 4 models (cumulative posterior probability = 1 ):
##
##      p!=0  EV      SD      model 1      model 2      model 3      model 4
## Intercept 100  0.76092 1.230e+00      1.45068      0.99831      -1.05711      -0.79000
## age        0.0  0.00000 0.000e+00      .          .          .          .
## lwt        74.4 -0.01306 9.639e-03      -0.01865      -0.01406      .          .
## race       0.0
## .2         .    0.00000 0.000e+00      .          .          .          .
## .3         .    0.00000 0.000e+00      .          .          .          .
## smoke      0.0
## .1         .    0.00000 0.000e+00      .          .          .          .
## ptl        13.3
## .1         .    0.23225 6.179e-01      .          .          1.75026      .
## .2         .    0.08647 4.047e-01      .          .          0.65165      .
## .3         .    -1.79255 3.216e+02      .          .          -13.50896      .
## ht         56.6
## .TRUE      .    1.04938 1.060e+00      1.85551      .          .          .
## ui         0.0
## .1         .    0.00000 0.000e+00      .          .          .          .
## ftv        0.0  0.00000 0.000e+00      .          .          .          .
##
##      nVar
## BIC      -753.82285  -751.51602  -750.92334  -750.77644
## post prob      0.566      0.178      0.133      0.123
```

Models selected by BMA



Predictions

```

npkBMA = bicreg(x = npk[, c("block", "N", "P", "K")], y = npk$yield)
summary(npkBMA)

##
## Call:
## bicreg(x = npk[, c("block", "N", "P", "K")], y = npk$yield)
##
##
## 31 models were selected
## Best 5 models (cumulative posterior probability = 0.4714 ):
##
##          p!=0    EV      SD      model 1  model 2  model 3  model 4  model 5
## Intercept 100.0  53.5651  2.6232   55.125   50.742   54.371   56.333   55.717
## block2    47.0   2.0923  2.9592      .        5.892    2.263      .        .
## block3    88.6   5.9034  3.4860    4.833    9.217    5.588      .        4.833
## block4    67.0  -3.5910  3.3608   -5.817      .   -5.062   -7.025   -5.817
## block5    60.9  -3.1264  3.2565   -5.417      .   -4.663   -6.625   -5.417
## block6    34.4   1.2048  2.4195      .        4.792      .        .        .
## N1        100.0   5.6167  1.6870    5.617    5.617    5.617    5.617    5.617
## P1         15.6  -0.1845  0.7817      .        .        .        .   -1.183
## K1         91.4  -3.6402  1.9466   -3.983   -3.983   -3.983   -3.983   -3.983
##
## nVar              5          5          6          4          6
## r2                0.688    0.674    0.704    0.608    0.698
## BIC               -12.097  -11.034  -10.150   -9.792   -9.669
## post prob         0.183    0.107    0.069    0.058    0.054

```

Predictions

```
p = predict(npkBMA, newdata = npk, quantiles = c(.5,.05,.95))
bind_cols(npk, as.data.frame(p$quantiles)) |> head(20)
```

##	block	N	P	K	yield	0.5	0.05	0.95
## 1	1	0	1	1	49.5	49.74860	40.96947	58.48361
## 2	1	1	1	0	62.8	59.01439	50.23957	67.69668
## 3	1	0	0	0	46.8	53.58095	44.77569	62.30043
## 4	1	1	0	1	57.0	55.54910	46.77694	64.28062
## 5	2	1	0	0	59.8	61.29444	53.15264	69.32592
## 6	2	1	1	1	58.5	57.45633	49.44848	65.42687
## 7	2	0	0	1	55.5	52.02752	44.00540	59.99400
## 8	2	0	1	0	56.0	55.48986	47.40503	63.48321
## 9	3	0	1	0	62.8	57.00405	47.21705	66.71617
## 10	3	1	1	1	55.8	58.97445	49.33077	68.57627
## 11	3	1	0	0	69.5	62.80984	52.96439	72.55522
## 12	3	0	0	1	55.0	53.54594	43.88129	63.14868
## 13	4	1	0	0	62.0	58.34556	48.84541	67.82723
## 14	4	1	1	1	48.8	54.52696	44.85351	64.14974
## 15	4	0	0	1	45.5	49.09411	39.42632	58.71447
## 16	4	0	1	0	44.2	52.54720	43.07168	61.98942
## 17	5	1	1	0	52.0	58.31376	49.07504	67.54200
## 18	5	0	0	0	51.5	52.88006	43.60689	62.15077
## 19	5	1	0	1	49.8	54.85841	45.45391	64.25045
## 20	5	0	1	1	48.8	49.05686	39.65504	58.44869

MCMC for sampling θ and M

Suppose, you construct a Markov chain to sample jointly from $p(M_\gamma, \theta_\gamma | y)$. An issue here is that when you move $M_\gamma \rightarrow M_{\gamma'}$, there is a chance that you change the dimension of θ , i.e. $\sum_{i=1}^P \gamma_i \neq \sum_{i=1}^P \gamma'_i$. This can be done via Metropolis-Hastings where the change of dimension is taken into account and this approach is called **reversible jump MCMC**.

An alternative is to fully incorporate γ as a parameter in the model. For example,

$$\begin{aligned} y_{ij} &\stackrel{\text{ind}}{\sim} N(\gamma_i \theta_i, \sigma^2) \\ \theta_i &\stackrel{\text{ind}}{\sim} N(\mu, \tau) \\ \gamma_i &\stackrel{\text{ind}}{\sim} \text{Ber}(w_i) \end{aligned}$$

This is essentially a way to implement the point-mass prior.

MCMC for Model averaging for GLMs

We can implement a similar MCMC to perform model averaging in Bayesian GLMs. Let $\theta_i = E[y_i|\theta_i]$ and ϕ as a dispersion parameter, then we can define a GLM as

$$\begin{aligned} y_i &\sim p(y_i|\theta_i, \psi) \\ \theta_i &= g^{-1}(X_i'\beta) \\ \beta_j &= \gamma_j\phi_j \\ \phi_j &\stackrel{ind}{\sim} N(\mu, \tau) \\ \gamma_j &\stackrel{ind}{\sim} Ber(w_j) \end{aligned}$$

For probit and ordinal regression, we can augment the model with parameters ζ_i , e.g. for probit regression

$$y_i = I(\zeta_i > 0) \text{ and } \zeta_i \stackrel{ind}{\sim} N(\theta_i, \psi).$$

There is a similar augmentation for logistic regression, see the BayesLogit R package and references therein. For these models and linear regression, we can construct an MCMC entirely using Gibbs steps. For other models, e.g. Poisson regression, sampling ϕ_j results in a non-Gibbs step.

BMS

```
library(BMS) # Bayesian model sampling
data(datafls)
dim(datafls)

## [1] 72 42

bma1 = bms(
  X.data = datafls,
  burn   = 1000,
  iter   = 10000,
  g       = "EBL",          # Model specific empirical Bayes estimate
  mprior  = "uniform",      # model over priors (extremely flexible)
  user.int = interactive())
```



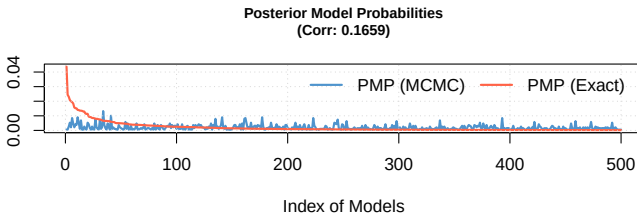
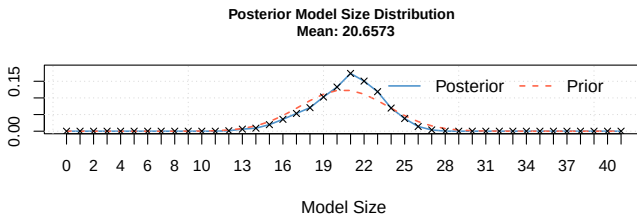
```
print(bma1)
```

##		PIP	Post Mean	Post SD	Cond.Pos.Sign	Idx
##	GDP60	1.0000	-1.623019e-02	3.195945e-03	0.00000000	12
##	Confucian	1.0000	6.025620e-02	1.367022e-02	1.00000000	19
##	LifeExp	0.9943	8.640151e-04	2.602475e-04	1.00000000	11
##	SubSahara	0.9629	-1.830845e-02	7.039820e-03	0.00000000	7
##	Hindu	0.9025	-6.232275e-02	3.541325e-02	0.00277008	21
##	EquipInv	0.8871	1.102536e-01	5.953501e-02	1.00000000	38
##	Mining	0.8814	3.389265e-02	1.887838e-02	1.00000000	13
##	NequipInv	0.8557	4.421700e-02	2.705937e-02	1.00000000	39
##	LabForce	0.8294	2.064558e-07	1.382417e-07	0.98312033	29
##	EcoOrg	0.7516	1.512516e-03	1.174784e-03	1.00000000	14
##	RuleofLaw	0.7282	8.304831e-03	6.773861e-03	0.99945070	26
##	LatAmerica	0.7045	-7.850392e-03	7.016886e-03	0.00411639	6
##	HighEnroll	0.6526	-5.488648e-02	5.276091e-02	0.00781489	30
##	EthnoL	0.6238	6.554575e-03	6.603552e-03	0.99983969	20
##	Protestants	0.6125	-6.063387e-03	6.265295e-03	0.00000000	25
##	BlMktPm	0.5190	-3.692555e-03	4.399196e-03	0.00000000	41
##	Spanish	0.4857	4.737145e-03	6.571061e-03	0.97199918	2
##	French	0.4812	3.389677e-03	4.853706e-03	0.98877805	3
##	Muslim	0.4438	4.660786e-03	6.791121e-03	0.99616945	23
##	PrScEnroll	0.4080	5.740752e-03	9.319934e-03	0.98578431	10
##	PolRights	0.3983	-6.207127e-04	1.079525e-03	0.02912378	33
##	WarDummy	0.3838	-1.219759e-03	2.174250e-03	0.00000000	5
##	OutwarOr	0.3837	-1.102930e-03	1.942896e-03	0.01485536	8
##	CivlLib	0.3787	-7.136884e-04	1.264790e-03	0.02587800	34
##	Age	0.3693	-1.408812e-05	2.522040e-05	0.00270783	16
##	Buddha	0.3575	3.353475e-03	5.973840e-03	0.99748252	17
##	Brit	0.3507	1.336366e-03	3.346241e-03	0.76846307	4
##	YrsOpen	0.3385	1.887588e-03	4.403748e-03	0.90576071	15
##	English	0.3303	-2.348280e-03	4.365967e-03	0.00000000	35
##	PublEduPct	0.3191	4.996191e-02	1.090692e-01	0.92760890	31
##	Area	0.2518	-4.415916e-08	3.647510e-07	0.33876092	9
##	PrExports	0.2495	-1.282406e-03	4.043301e-03	0.09458918	24

```
summary(bma1)
```

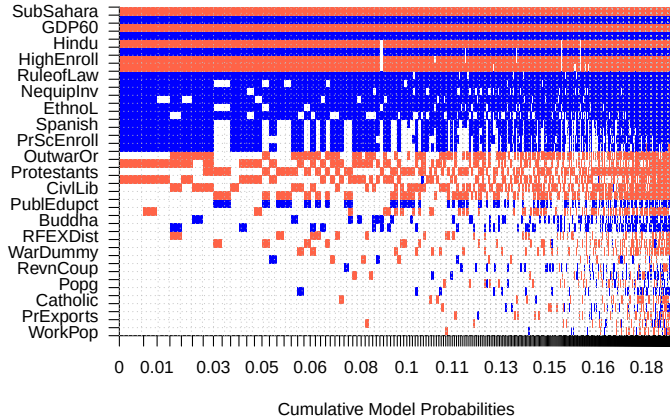
## Mean no. regressors	Draws	Burnins	Time	No. models visited
## "20.6573"	"10000"	"1000"	"0.4739559 secs"	"4446"
## Modelspace 2^K	% visited	% Topmodels	Corr PMP	No. Obs.
## "2.2e+12"	"2e-07"	"19"	"0.1659"	"72"
## Model Prior	g-Prior	Shrinkage-Stats		
## "uniform / 20.5"	"EBL"	"Av=0.9611"		

```
plot(bma1)
```

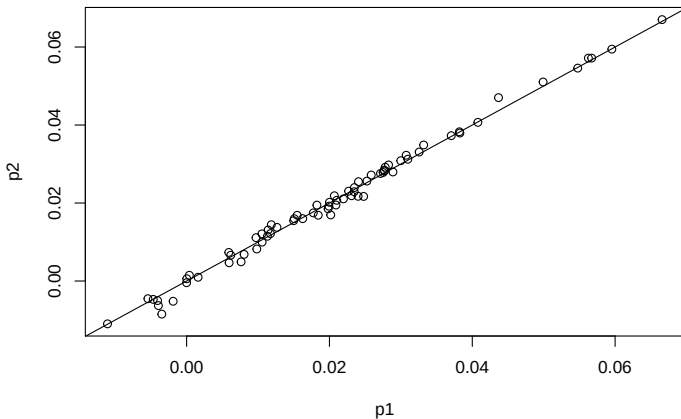


```
image(bma1)
```

Model Inclusion Based on Best 500 Models



```
p1 = predict(bma1)           # fitted values based on MCMC frequencies
p2 = predict(bma1, exact=TRUE) # fitted values based on best models
plot(p1,p2); abline(0,1)
```



Model averaging of regression coefficients?

```

set.seed(20171007)
n <- 100
x1 <- rnorm(n)
y <- x1 + rnorm(n)
x2 <- x1 + rnorm(n,0,.01)

library(BMA)
m <- bicreg(cbind(x1,x2), y)

summary(m)

##
## Call:
## bicreg(x = cbind(x1, x2), y = y)
##
##
##      3 models were selected
## Best 3 models (cumulative posterior probability = 1 ):
##
##      p!=0    EV      SD      model 1    model 2    model 3
## Intercept 100.0  0.09687  0.08774    0.09686    0.09646    0.09980
## x1         51.6  -0.12069  3.07643      .         0.78988   -7.23838
## x2         54.9   0.91107  3.07714    0.79065      .         8.03022
##
## nVar              1              1              2
## r2                0.480          0.479          0.483
## BIC               -60.81432      -60.67277      -56.82626
## post prob         0.484          0.451          0.066

```